Homework #1

prob 1.)

Givens:

- $\vec{\nabla}_{cart} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$
- $\vec{\nabla}_{cylc} = \hat{\rho}\frac{\partial}{\partial \rho} + \hat{\phi}\frac{1}{\rho}\frac{\partial}{\partial \phi} + \hat{z}\frac{\partial}{\partial z}$
- $\vec{\nabla}_{sphrc} = \hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}$

objective:

a.) take gradient of f in cartesian coordinate

b.) take gradient of f in cylindrical coordinate

c.) take gradient of f in spherical coord

a.) $f = x^2y^3z^4$

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$\frac{\partial f}{\partial x} = 2xy^3z^4$ $\quad \frac{\partial f}{\partial y} = 3x^2y^2z^4$ $\quad \frac{\partial f}{\partial z} = 4x^2y^3z^3$

$$\vec{\nabla}f = 2xy^3z^4\,\hat{\imath} + 3x^2y^2z^4\,\hat{\jmath} + 4x^2y^3z^3\,\hat{k}$$

b) $f = \cos\phi \frac{1}{\rho}$

$\vec{\nabla}_{cyl} = \hat{\rho}\frac{\partial}{\partial \rho} + \hat{\phi}\frac{1}{\rho}\frac{\partial}{\partial \phi} + \hat{z}\frac{\partial}{\partial z}$

- $\frac{\partial f}{\partial \rho} = \cos\phi \frac{\partial}{\partial \rho}\left(\frac{1}{\rho}\right) = \cos\phi \frac{\partial}{\partial \rho}\left(\rho^{-1}\right)$

  $= \cos\phi\left(-1\rho^{-2}\right) = -\cos\phi\left(\frac{1}{\rho^2}\right)$

- $\frac{\partial f}{\partial \rho} = \frac{-\cos\phi}{\rho^2}$

- $\frac{\partial f}{\partial \phi} = \frac{1}{\rho}\frac{\partial}{\partial \phi}(\cos\phi)$

  $= \frac{1}{\rho}(-\sin\phi) = \frac{-\sin\phi}{\rho}$

- $\frac{\partial f}{\partial z} = 0$

$$\vec{\nabla} f = -\frac{\cos\phi}{\rho^3}\hat{\rho} - \frac{\sin\phi}{\rho}\hat{\phi}$$

c.) • $f = r\cos\theta$

• $\vec{\nabla}_{sphere} = \hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}$

• $\vec{\nabla} f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$

• $\frac{\partial f}{\partial r} = \cos\theta$, $\frac{\partial f}{\partial \theta} = -r\sin\theta$

• $\frac{\partial f}{\partial \phi} = 0$

• $\vec{\nabla} f = \hat{r}\cos\theta + \hat{\theta}\frac{1}{r}(-r\sin\theta) + \hat{\phi}\frac{1}{r\sin\theta}(0)$

• $\boxed{\vec{\nabla} f = \cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}}$

prob 2.)

objective:

a & b.) find divergence and curl.

Givens:

- Cartesian:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- $$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A_x & A_y & A_z \end{vmatrix}$$

- spherical:

$$\vec{\nabla} \cdot \vec{V} = \frac{1}{r^2} \frac{\partial (r^2 V_r)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial (\sin\theta V_\theta)}{\partial \theta} + \frac{1}{r \sin\theta} \frac{\partial V_\phi}{\partial \phi}$$

$$\vec{\nabla} \times \vec{V} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \partial/\partial u_1 & \partial/\partial u_2 & \partial/\partial u_3 \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}$$

- $u_1 = r, \quad u_2 = \theta, \quad u_3 = \phi$

$$\vec{r} = r\sin\theta\cos\phi\,\hat{\imath} + r\sin\theta\sin\phi\,\hat{\jmath} + r\cos\theta\,\hat{k}$$

- $$\frac{\partial \vec{r}}{\partial r} = \sin\theta\cos\phi\,\hat{\imath} + \sin\theta\sin\phi\,\hat{\jmath} + \cos\theta\,\hat{k}$$

- $$h_1 = \left|\frac{\partial \vec{r}}{\partial r}\right| = \sqrt{(\sin\theta\cos\phi)^2 + (\sin\theta\sin\phi)^2 + (}$$

$$= \sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}$$

$$= \sqrt{\sin^2\theta\left(\cos^2\phi + \sin^2\phi\right) + \cos^2\theta} = |$$

$$h_1 = |$$

$$\frac{\partial \vec{r}}{\partial \theta} = r\cos\theta\cos\phi\,\hat{i} + r\cos\theta\sin\phi\,\hat{j} - r\sin\theta\,\hat{k}$$

$$h_2 = \sqrt{(r\cos\theta\cos\phi)^2 + (r\cos\theta\sin\phi)^2 + (r\sin\theta)^2}$$

$$= \sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}$$

$$= \sqrt{r^2\cos^2\theta(\cos^2\phi + \sin^2\phi) + r^2\sin^2\theta}$$

$$= \sqrt{r^2\cos^2\theta + r^2\sin^2\theta}$$

$$= r$$

$$\therefore h_2 = r$$

$$h_3 = r\sin\theta$$

$$\vec{\nabla} \times \vec{V} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\,\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ V_r & rV_\theta & r\sin\theta\, V_\phi \end{vmatrix}$$

a.) $\vec{f} = x^2 y^3 \hat{\imath} + z^4 \hat{k}$

$$\vec{\nabla} \cdot \vec{f} = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

$$= \frac{\partial}{\partial x}\left(x^2 y^3\right) + \frac{\partial}{\partial y}\left(0\right) + \frac{\partial}{\partial z}\left(z^4\right)$$

$$= 2xy^3 + 4z^3$$

$$\boxed{\vec{\nabla} \cdot \vec{f} = 2xy^3 + 4z^3}$$

$$\vec{\nabla} \times \vec{f} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 y^3 & 0 & z^4 \end{vmatrix}$$

$$\hat{i}\left(\frac{\partial}{\partial y}\left(z^4\right)\right) - \hat{j}\left(\frac{\partial}{\partial x}\left(z^4\right) - \frac{\partial}{\partial z}\left(x^2 y^3\right)\right)$$

$$+ \hat{k}\left(0 - \frac{\partial}{\partial y}\left(x^2 y^3\right)\right)$$

$$\vec{\nabla} \times \vec{f} = \hat{k}\left(-3y^2 x^2\right) = -3x^2 y^2 \hat{k}$$

b.) $\vec{f} = \dfrac{\hat{r}}{r^3}$

$$\nabla \cdot \vec{f} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{1}{r}\right) = \frac{1}{r}\left(-r^{-2}\right)$$

$$= \frac{1}{r}\left(\frac{-1}{r^2}\right) = \frac{-1}{r^3}$$

$$\boxed{\nabla \cdot \vec{f} = \frac{-1}{r^3}}$$

$$\nabla \times \vec{f} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\,\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ \frac{1}{r^3} & 0 & 0 \end{vmatrix}$$

$$= \frac{1}{r^2\sin\theta}\left[\hat{r}(0) - r\hat{\theta}\left(0 - \frac{\partial}{\partial \phi}\left(\frac{1}{r^3}\right)\right) + r\sin\theta\,\hat{\phi}\left(0 - \frac{\partial}{\partial \theta}\left(\frac{1}{r^3}\right)\right)\right]$$

$$\boxed{\nabla \times \vec{f} = 0}$$

Prob 3.)

Objection:

Using charges and dirac delta functions find $\rho(\vec{x})$.

Problem outline:

1.) Define coordinate system, charge distribution relation, Total charge equation, where the particle is at.

a.) Define coordinate system

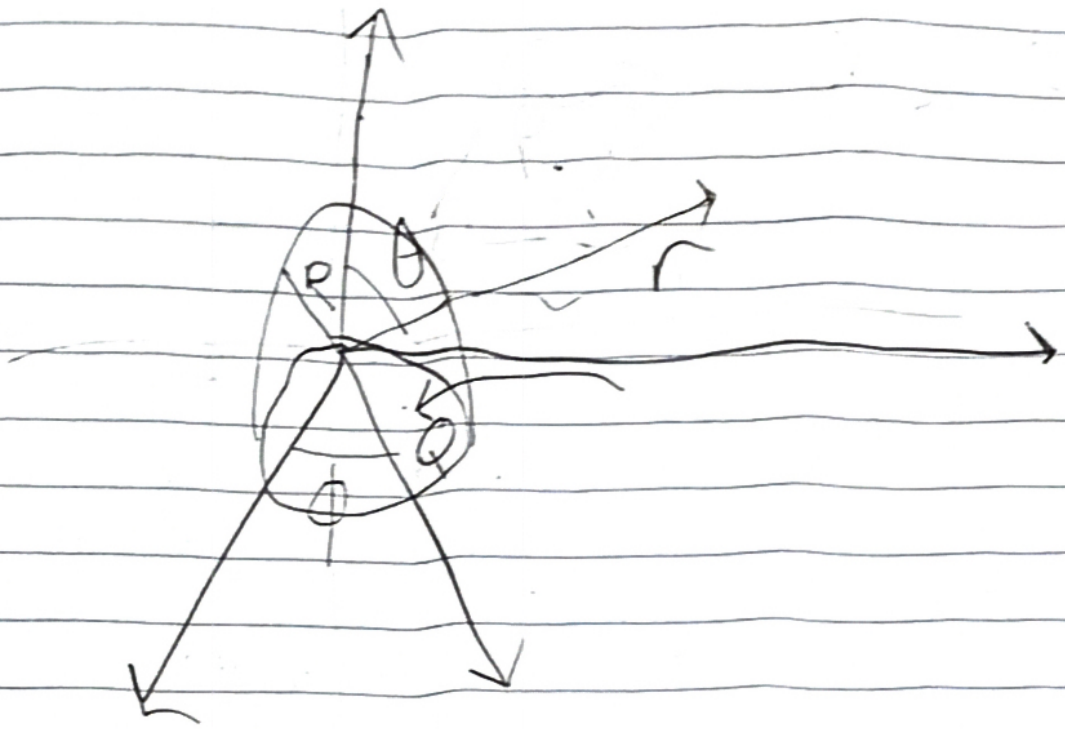
b.) $\rho = A\delta(r)$ define

c.) $Q = \int \rho \, dl, ds \text{ or } dv$ (define integration limits.)

d.) Integrate and find constant. (Volumes)

a.)

- Spherical coordinates



- Current density volume distribution

$$\rho = A\,\delta(r-R) \quad , Q = \int_V \rho\, dV$$

- find this constant.

- $d^3x = r^2 \sin\theta\, dr\, d\theta\, d\phi$
- $0 \le r \le \infty \qquad 0 \le \phi \le 2\pi$,
  $0 \le \theta \le \pi$.

- $$Q = \int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} r^2 \sin\theta\, dr\, d\theta\, d\phi\; A\,\delta(r-R)$$

$$Q = A\int_0^{\infty} r^2\delta(r-R)dr \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta\, d\theta$$

- $\int_0^{2\pi} d\phi = 2\pi$, $\int_0^{\pi} \sin\theta d\theta = -\cos\theta\Big|_0^{\pi}$

$= 1+1 = 2$

- $A\int_0^{\infty} r^2\delta(r-R)dr$ • R is inside Limits

$= AR^2$

$Q = AR^2(4\pi)$

$A = \frac{Q}{4\pi R^2}$

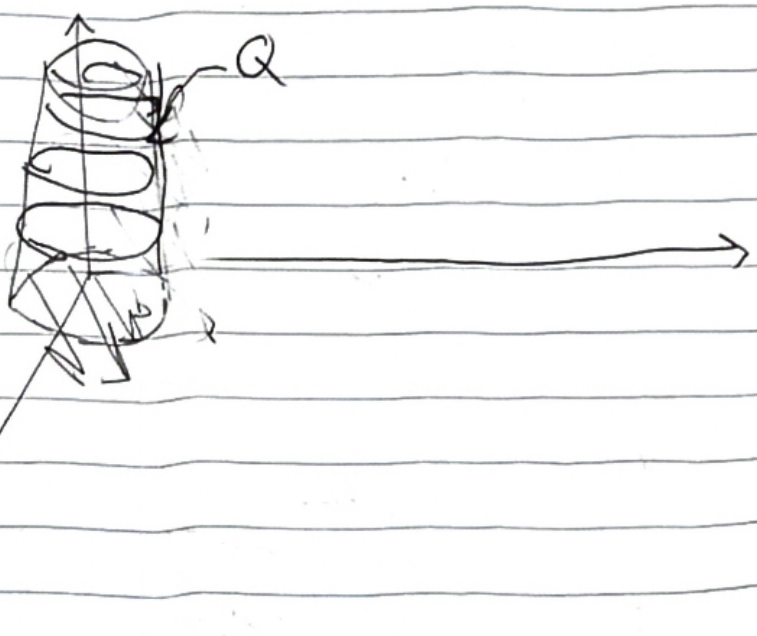
$$\boxed{\rho = \frac{Q}{4\pi R^2}\delta(r-R)}$$

b.)

- Cylindrical coordinates.
- $\sigma \rightarrow$ surface charge density
- infinite cylinder with radius b.
- $\lambda \rightarrow$ charge on lengths of the cylinder.

- $\sigma = A\delta(r-b)$
-

Cylindrical coordinates (r, θ, z)



- $ds = r\,dr\,d\theta$
- Cross-section $\quad 0 \le \theta \le 2\pi$, $\quad 0 \le r \le \infty$

$$\lambda = A\int_0^{\infty} r\delta(r-b)\,dr \int_0^{2\pi} d\theta$$

$$q = A\int_0^{\infty} r\,\delta(r-b)\,dr \int_0^{2\pi} d\theta$$

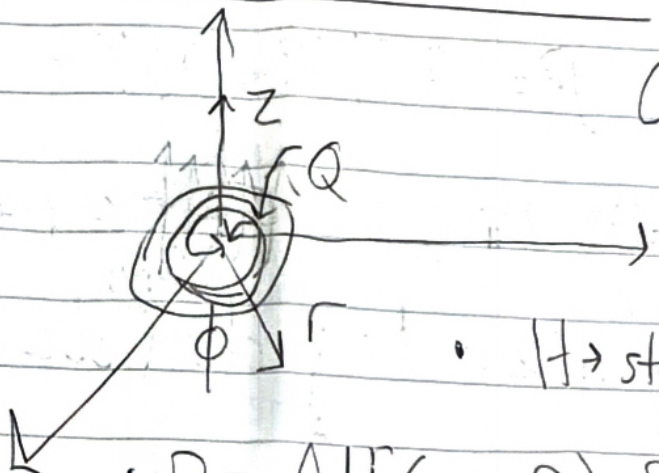
$$\cdot \int_0^{2\pi} d\theta = 2\pi$$

$$\cdot A\int_0^{\infty} r\delta(r-b)\,dr \qquad r = b$$
$$= Ab$$

$$\lambda = Ab\,2\pi$$

$$b = \frac{\lambda}{2\pi b}, \quad \boxed{\rho = \frac{\lambda}{2\pi b}\delta(r-b)}$$

C) Cylindrical coordinates:

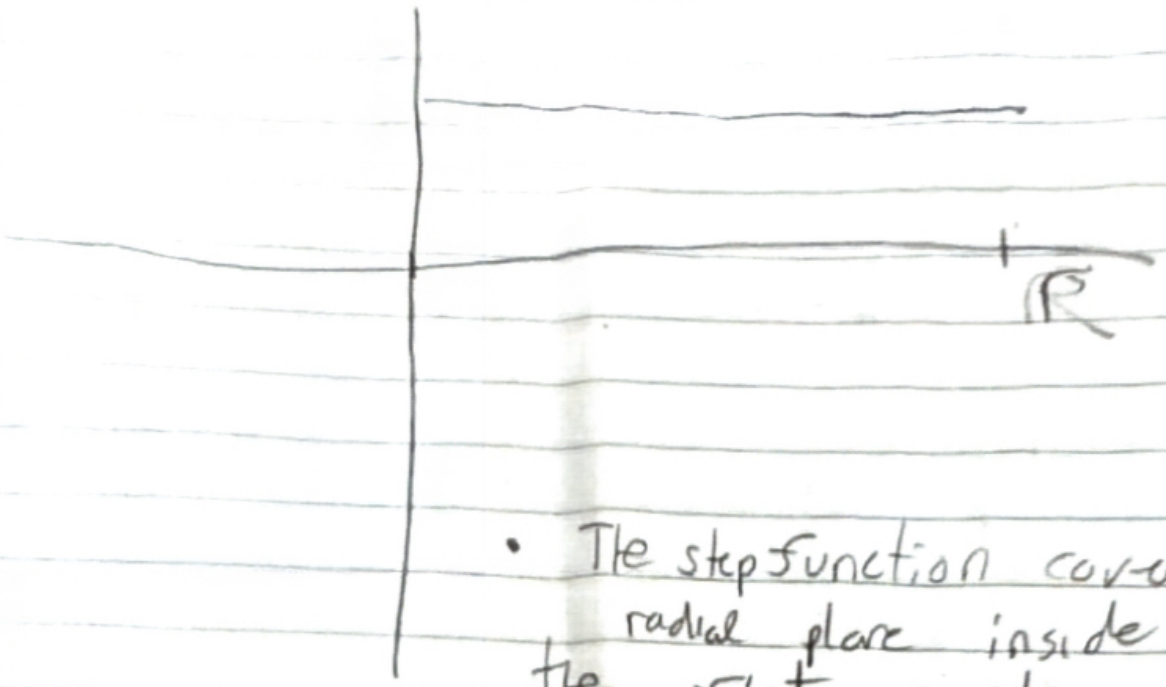


(r, θ, z)

· H → step function

· $\rho = AH(r-R)\,\delta(z)$

$\Pi(R-r) = \begin{cases} 1 & R \le r \\ 0 & R > r \end{cases}$



- The step function covers radial plane inside the flat circle.

- $\rho = A\,H(R-r)\,\delta(z)$, $ds = r\,dr\,d\theta\,dz$
- $0 \le \theta \le 2\pi$, $0 \le r \le R+\epsilon$, $-\infty \le z \le \infty$

$$Q = A\int_0^{R+\epsilon} r\,H(R-r)\,dr \int_0^{2\pi} d\theta \int_{-\infty}^{\infty} dz\,\delta(z)$$

- $Q = A\,\frac{r^2}{2}\Big|_0^{R+\epsilon}(2\pi)(1)$

- $\lim_{\epsilon \to 0} Q = \lim_{\epsilon \to 0} \frac{A}{2}(2\pi)\left(R^2 + 2\epsilon R + \epsilon^2\right)$

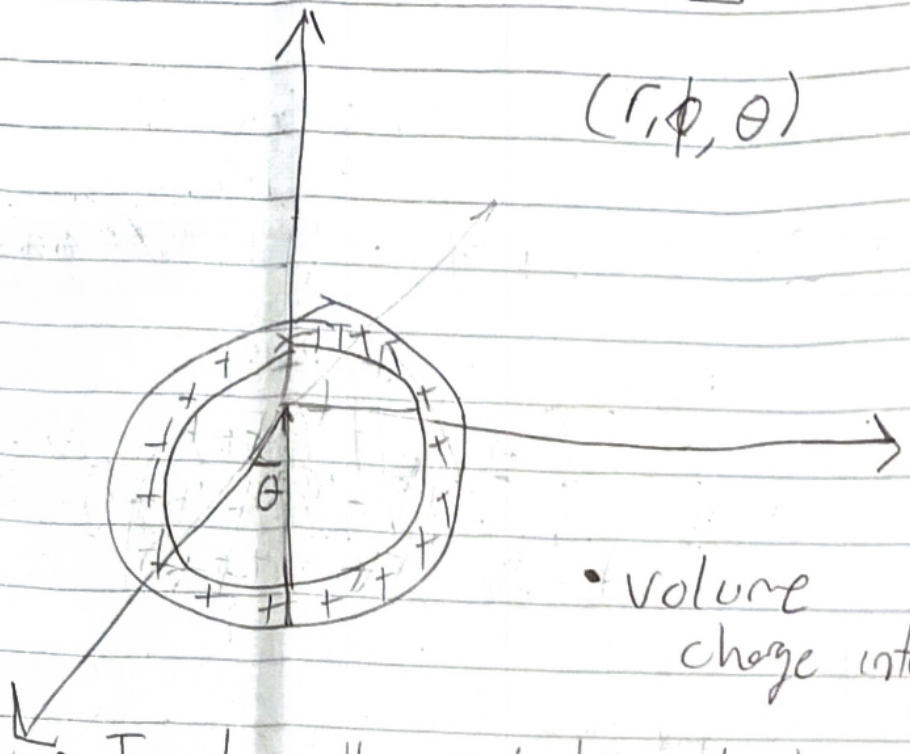
- $Q = A\pi R^2$

- $\frac{Q}{\pi R^2} = A$

$$\rho = \frac{Q}{\pi R^2} H(R-r)\,\delta(z)$$

d.)

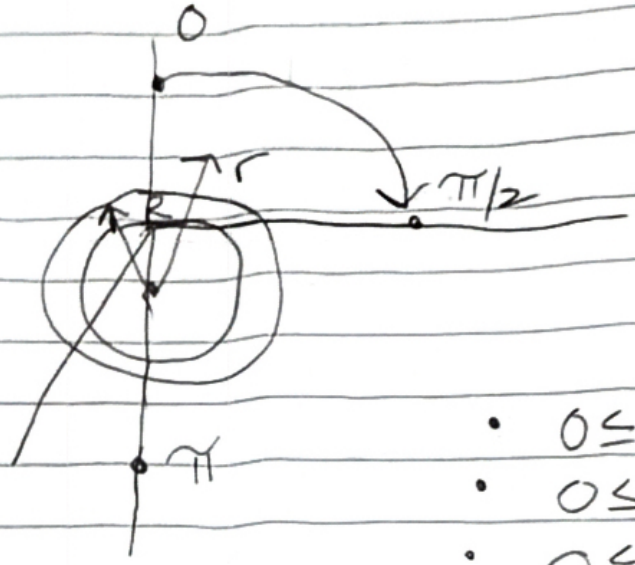
(r, ϕ, θ)



- volume charge integration

- I need math property $\frac{1}{r}$ coordinate that describes charge distributed over whole Ring and that the ring is in a particular plane.

$$\rho = A\,\delta(R-r)\,\delta\left(\theta-\frac{\pi}{2}\right)$$



- $0 \le r \le \infty$
- $0 \le \theta \le \pi$
- $0 \le \phi \le 2\pi$

- $dV = r^2 \sin\theta\, dr\, d\theta\, d\phi$

- $$Q = \int_{\text{all space}} \rho\, dV = \int_0^{\infty}\int_0^{\pi}\int_0^{2\pi} A r^2 \sin\theta\,\delta(R-r)\,\delta\left(\theta-\frac{\pi}{2}\right) dr\, d\theta\, d\phi$$

- $$Q = A\int_0^{\infty} r^2\delta(R-r)\,dr \int_0^{\pi} \sin\theta\,\delta\left(\theta-\frac{\pi}{2}\right) d\theta \int_0^{2\pi} d\phi$$

- $$A\left(R^2\right)\left(\sin\left(\frac{\pi}{2}\right)\right)(2\pi) = A R^2 2\pi$$

$$A = \frac{Q}{2\pi R^2}$$

$$\rho = \frac{Q}{2\pi R^2}\,\delta(R-r)\,\delta(\theta-\pi/2)$$

Prob 4)

Given:

- $\Phi = \frac{q}{4\pi\varepsilon_0} \frac{e^{-\alpha r}}{r}\left(1 + \frac{\alpha r}{2}\right)$
- $r \rightarrow$ radius in spherical coordinates
- assume we have a sphere.

Objective:

- Find ρ both continuous and discrete
- Interpret the physics.

Gauss's Law:

$$\nabla^2 \phi(\vec{r}) = -\frac{\rho(\vec{r})}{\varepsilon_0}$$

$$k = \frac{q}{4\pi\varepsilon_0}$$

$$\phi = \frac{k e^{-\alpha r}}{r} + \frac{k\alpha e^{-\alpha r}}{2}$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right)$$

- $\phi = \frac{ke^{-\alpha r}}{r} + \frac{k\alpha e^{-\alpha r}}{2}$
- $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right)$
- $\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{ke^{-\alpha r}}{r} + \frac{k\alpha e^{-\alpha r}}{2}\right)\right)$
  - $\frac{\partial}{\partial r}\left(\frac{ke^{-\alpha r}}{r}\right) = \frac{\partial}{\partial r}\left(ke^{-\alpha r}\right)\frac{1}{r} + \frac{\partial}{\partial r}\left(\frac{1}{r}\right)ke^{-\alpha r}$
    $= \frac{-k\alpha e^{-\alpha r}}{r} + ke^{-\alpha r}\frac{\partial}{\partial r}\left(\frac{1}{r}\right)$
  - $\frac{\partial}{\partial r}\left(\frac{k\alpha e^{-\alpha r}}{2}\right) = -\frac{k\alpha^2 e^{-\alpha r}}{2}$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\left(\frac{-k\alpha e^{-\alpha r}}{r} + ke^{-\alpha r}\frac{\partial}{\partial r}\left(\frac{1}{r}\right) - \frac{k\alpha^2 e^{-\alpha r}}{2}\right)\right)$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(-k\alpha r e^{-\alpha r} + ke^{-\alpha r}r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right) - \frac{k\alpha^2 r^2 e^{-\alpha r}}{2}\right)$$

$$= \frac{1}{r^2}\left(-k\alpha e^{-\alpha r} + k\alpha^2 r e^{-\alpha r} - \alpha k e^{-\alpha r}r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right) + ke^{-\alpha r}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right) + \frac{k\alpha^3 e^{-\alpha r}r^2}{2} - k\alpha^2 r e^{-\alpha r}\right)$$

- $\frac{\partial}{\partial r}\left(\frac{1}{r}\right) \to \frac{-1}{r^2}$

$$= \left(\frac{-k\alpha e^{-\alpha r}}{r^2} + \frac{k\alpha^2 e^{-\alpha r}}{r} - \alpha k e^{-\alpha r}\frac{\partial}{\partial r}\left(\frac{1}{r}\right) + k e^{-\alpha r}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right) \right.$$

$$\left. + \frac{k\alpha^3 e^{-\alpha r}}{2} - \frac{k\alpha^2 e^{-\alpha r}}{r} \right)$$

$$= \left(\frac{-k\alpha e^{-\alpha r}}{r^2} + \frac{k\alpha^2 e^{-\alpha r}}{r} + \frac{\alpha k e^{-\alpha r}}{r^2} + k e^{-\alpha r}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right) \right.$$

$$\left. + \frac{k\alpha^3 e^{-\alpha r}}{2} - \frac{k\alpha^2 e^{-\alpha r}}{r} \right)$$

- $-\frac{\rho}{\varepsilon_0} = k e^{-\alpha r}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right) + \frac{k\alpha^3 e^{-\alpha r}}{2}$

- $\rho = -\varepsilon_0 k e^{-\alpha r}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\left(\frac{1}{r}\right)\right) - \frac{k\alpha^3\varepsilon_0 e^{-\alpha r}}{2}$

- $\varepsilon_0 k = \frac{q}{4\pi}$, $\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta(r)$

- $\rho(\vec{r}) = \frac{-q e^{-\alpha r}\nabla^2\left(\frac{1}{r}\right)}{4\pi} - \frac{q\alpha^3 e^{-\alpha r}}{8\pi}$

- $\rho(\vec{r}) = q e^{-\alpha r}\delta(r) - \frac{q\alpha^3 e^{-\alpha r}}{8\pi}$

- Small approx: $r \to 0$
  - $\rho(\vec{r}) = q\delta(r)$
- Large approx: $r \to \infty$
  - $\rho(\vec{r}) = \frac{-q\alpha^3}{8\pi} e^{-\alpha r}$

$$\rho(\vec{r}) = \frac{-q\alpha^3 e^{-\alpha r}}{8\pi} + q\delta(r)$$

- Small approximation of charge can represent a point charge at the ground state or at the origin.
- Large approximation of charge represents a charge at a specific orbit/excited state that rapidly decays

prob 5.)

objective:

a.) a 3, b.) Show an electrostatic field. 3? Find Φ.

- $\vec{E}(x,y,z) = c\sqrt{x^2-y^2+2z^2}\,\hat{z}$

- To prove the electrostatic field:

- $\vec{\nabla} \times \vec{E}(\vec{r}) = 0$

- $\vec{\nabla} = \frac{\partial}{\partial x}\uparrow + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial y}\hat{k}$

- $\vec{\nabla} \times \vec{E}(\vec{r}) = \begin{vmatrix} \uparrow & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & c\sqrt{x^2-y^2+2z^2} \end{vmatrix}$

- $\vec{\nabla} \times \vec{E} = \hat{\imath}\left(\frac{\partial}{\partial y}\left(c\sqrt{x^2-y^2+2z^2}\right)\right)$

$-\hat{\jmath}\left(\frac{\partial}{\partial x}\left(c\sqrt{x^2-y^2+2z^2}\right)\right) + \hat{k}(0)$

- $= \hat{\imath}\left(C\frac{1}{2}\left(x^2-y^2+2z^2\right)^{-1/2}(-2y)\right)$

$-\hat{\jmath}\left(c\frac{1}{2}\left(x^2-y^2+2z^2\right)^{-1/2}(2x)\right)$

- $\vec{\nabla} \times \vec{E} = \hat{\imath}\left(-C\left(x^2-y^2+2z^2\right)^{-1/2} y\right) - \hat{\jmath}\left(C\left(x^2-y^2+2z^2\right)^{-1/2} x\right)$

- $\vec{\nabla} \times \vec{E} \neq 0$, So the electric field is not electrostatic.

$\cdot (r, \theta, \phi)$

b) $\cdot E(r,\theta,\phi) = C\left(r\cos^2\theta\,\hat{r} - r\cos\theta\sin\theta\,\hat{\theta}\right)$

$\cdot \vec{\nabla} = \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial\theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\hat{\phi}$

$\cdot h_1 = 1, \quad h_2 = r, \quad h_3 = r\sin\theta$

$\cdot \vec{\nabla}\times\vec{A} = \frac{1}{h_1 h_2 h_3}\begin{vmatrix} h_1\hat{e}_1 & h_2\hat{e}_2 & h_3\hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_r & A_\theta & A_\phi \end{vmatrix}$

$\cdot u_1 = r, \; u_2 = \theta, \; u_3 = \phi$

$\cdot \vec{\nabla}\times\vec{E} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\hat{\phi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ Cr\cos^2\theta & -Cr^2\cos\theta\sin\theta & 0 \end{vmatrix}$

$= \frac{1}{r^2\sin\theta}\left[\hat{r}\left(\frac{\partial}{\partial\theta}(0) - \frac{\partial}{\partial\phi}\left(-Cr^2\cos\theta\sin\theta\right)\right)\right.$

$$-r\hat{\theta}\left(\frac{\partial}{\partial r}\left(\quad 0 \quad\right)-\frac{\partial}{\partial \phi}\left(Cr\cos^2\theta\right)\right)$$

$$+\,r\sin\theta\,\hat{\phi}\left(\frac{\partial}{\partial r}\left(rCr^2\cos\theta\sin\theta\right)-\frac{\partial}{\partial \theta}\left(Cr\cos^2\theta\right)\right)$$

- $\vec{\nabla}\times\vec{E}=$

$$\frac{1}{r^2\sin\theta}\left[\cancel{r\sin\theta\,\hat{\phi}\left(-2rC\cos\theta\sin\theta+2Cr\cos\theta\sin\theta\right)}\right.$$

- $\vec{\nabla}\times\vec{E}=0$, so the E-field is an electrostatic field.

- $P(\vec{r})=-\int_{r_0}^{r}d\vec{l}\cdot\vec{E}$

- $d\vec{l}=dr\hat{r}+rd\theta\hat{\theta}+r\sin\theta d\phi\hat{\phi}$
- $\vec{E}=Cr\cos^2\theta\,\hat{r}-Cr\cos\theta\sin\theta\,\hat{\theta}$

- $\vec{E}\cdot d\vec{l}=Cr\cos^2\theta\,dr-Cr^2\cos\theta\sin\theta\,d\theta$

- $P(r,\theta) = -\int Cr\cos^2\theta\, dr + \int Cr^2\cos\theta\sin\theta\, d\theta$
- $P(r,\theta) = -C\cos^2\theta \int r\, dr + Cr^2 \int \cos\theta\sin\theta\, d\theta$
  - $-C\cos^2\theta \frac{r^2}{2} + D + Cr^2 \int \cos\theta\sin\theta\, d\theta$
    - $Cr^2 \int \cos\theta\sin\theta\, d\theta$

      $u = \sin\theta$

      $du = \cos\theta\, d\theta$
    - $Cr^2 \int u\, du$

      $= Cr^2 \frac{u^2}{2} = Cr^2 \frac{\sin^2\theta}{2}$
- $P(r,\theta) = -C\frac{\cos^2\theta r^2}{2} + C\frac{r^2\sin^2\theta}{2} + D$

  $r = 0, \quad P = 0$ (zero refrence)
- $D = 0$
- $\boxed{P(r,\theta) = -\frac{C\cos^2\theta r^2}{2} + \frac{Cr^2\sin^2\theta}{2}}$